

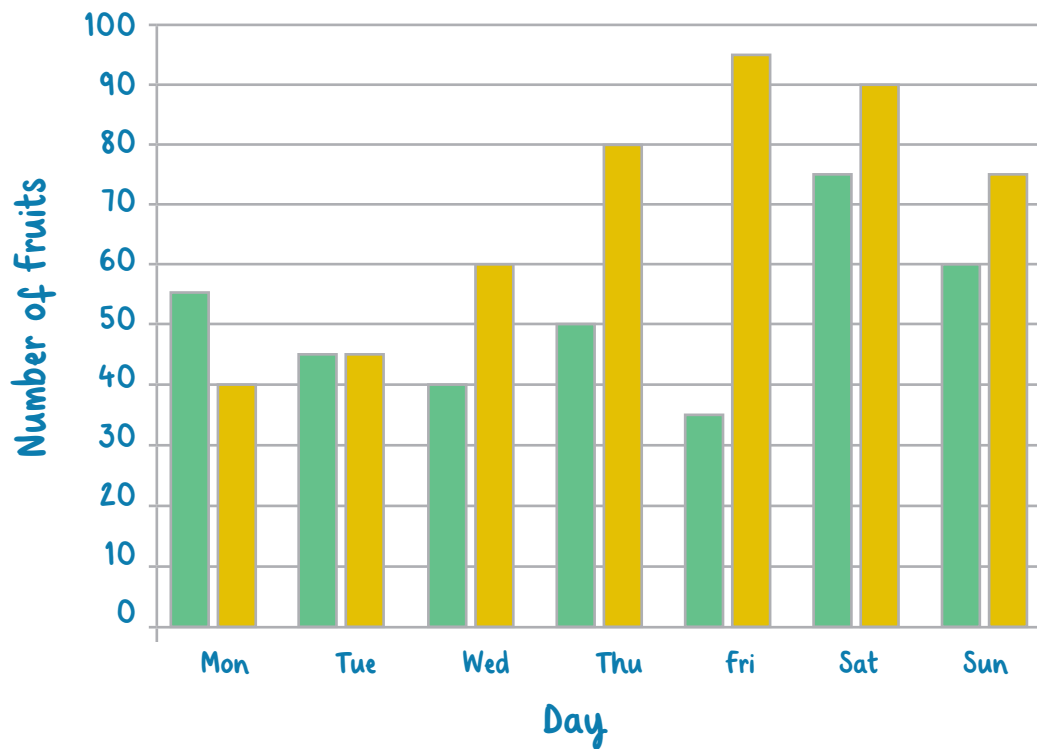
Statistics:

Bar Charts with Grouped Data Activities

The Fruit Stall

Frankie sells two different fruits each week on his fruit stall. Here are the results of his sales last week, when he was selling kiwis and bananas:

A Graph to Show the Number of Fruits Sold



Key	
	kiwis
	bananas

Use the data to answer the following questions.

1. On which day were the same amount of kiwis and bananas sold?

2. What is the difference between the amount of bananas sold and the amount of kiwis sold on Friday?

3. Frankie says, "I sold the most fruit on Friday."
Is he correct? Explain your answer.

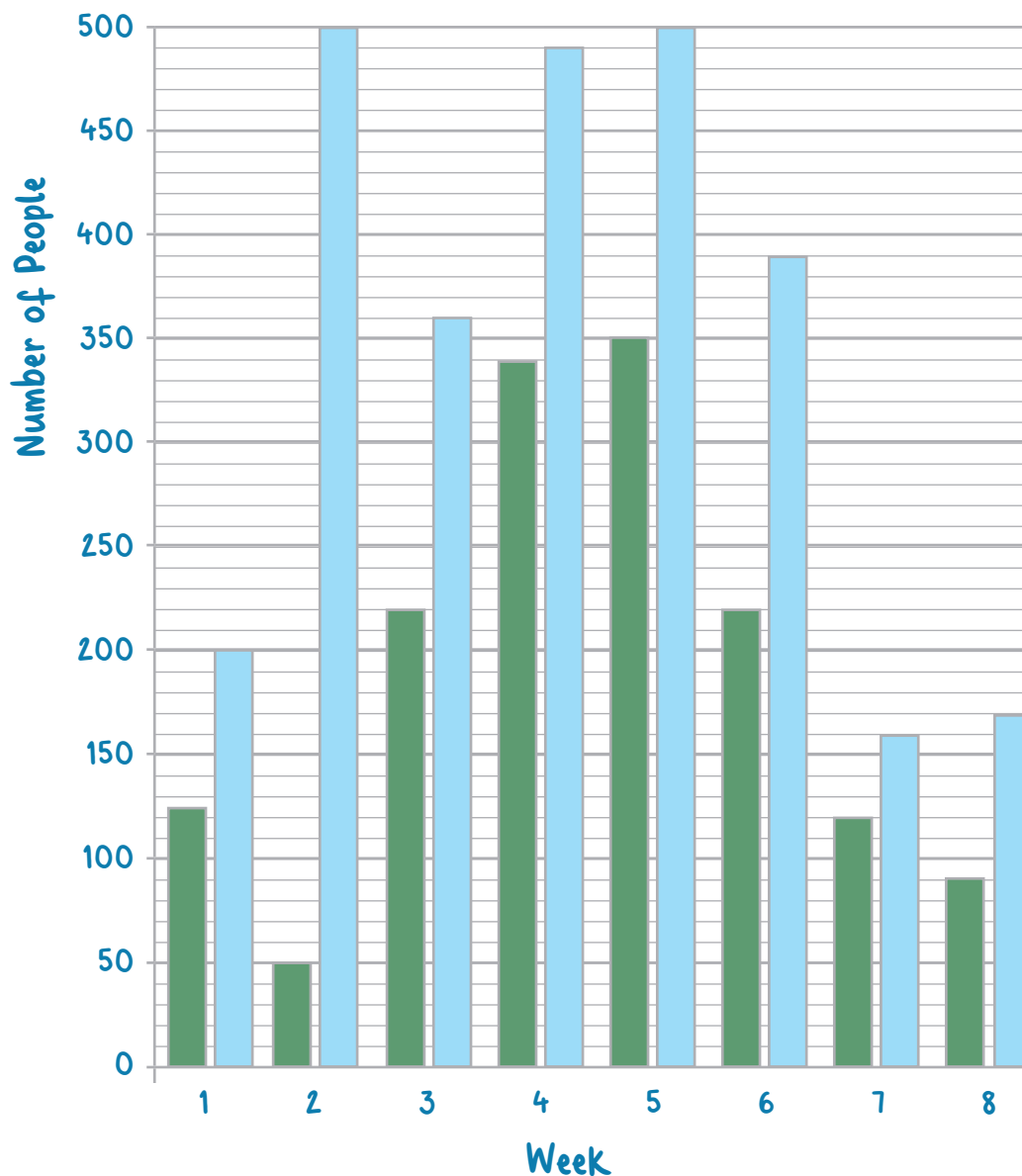
4. On what day were the fewest pieces of fruit sold?

5. Frankie says, "The amount of bananas I sold on Friday is equal to the amount of kiwis I sold on two different days."
Which two days could Frankie be talking about?

Pantomime Audiences

The local theatre has been keeping a record of the people in the audience for its Tuesday afternoon pantomime over the course of the eight week Christmas period. Here is the data.

A Graph to Show Audience Numbers over the Christmas Period on Tuesday Afternoons



Key	
	children
	adults

Use the data to answer the following questions.

1. The capacity of the theatre is 850. Complete the table to show many seats were empty on each performance.

Week								
Empty Seats								

2. Which two weeks do you think were the school holidays? Why?
-
3. The local school was invited to bring its children to a performance. Which performance do you think they brought them to? Why?
-
4. How many performances had over 100 fewer adults than children in the audience?
-
5. How many more children in total saw the Tuesday afternoon performances than adults?

Statistics: Bar Charts with Grouped Data – Answers

The Fruit Stall

1. On which day were the same amount of kiwis and bananas sold?

Tuesday

2. What is the difference between the amount of bananas sold and the amount of kiwis sold on Friday?

$$95 - 35 = 60$$

3. Frankie says, "I sold the most fruit on Friday."

Is he correct? Explain your answer.

Frankie is incorrect. On Friday, he sold 35 kiwis and 95 bananas, totalling 130.

However, the total on Saturday was $75 + 90 = 165$ and the total on Sunday was

$60 + 75 = 135$, so he sold more on two other days. The day he sold the most fruit was Saturday.

4. On what day were the fewest pieces of fruit sold?

Tuesday (90 pieces of fruit)

5. Frankie says, "The amount of bananas I sold on Friday is equal to the amount of kiwis I sold on two different days."

Which two days could Frankie be talking about?

He could be talking about:

- **Monday (55 kiwis) and Wednesday (40 kiwis)**
- **Friday (35 kiwis) and Sunday (60 kiwis)**
- **Tuesday (45 kiwis) and Thursday (50 kiwis)**

Pantomime Audiences

1. The capacity of the theatre is 850. Complete the table to show many seats were empty on each performance.

Week	1	2	3	4	5	6	7	8
Empty Seats	$125 + 200$ $= 325$	$50 + 500$ $= 550$	$220 + 360$ $= 580$	$340 + 490$ $= 830$	$350 + 500$ $= 850$	$220 + 390$ $= 610$	$120 + 160$ $= 280$	$90 + 170$ $= 260$
	$850 - 325$ $= 525$	$850 - 550$ $= 300$	$850 - 580$ $= 270$	$850 - 830$ $= 20$	$850 - 850$ $= 0$	$850 - 610$ $= 240$	$850 - 280$ $= 570$	$850 - 260$ $= 590$

2. Which two weeks do you think were the school holidays? Why?
Weeks 4 and 5 were the school holidays. The audiences were much bigger for these performances.
3. The local school was invited to bring its children to a performance. Which performance do you think they brought them to? Why?
The local school brought its children to the performance in week 2. The amount of adults was very small and the amount of children was large, which replicates the type of ratio of teachers to pupils.
4. How many performances had over 100 fewer adults than children in the audience?
5 (weeks 2, 3, 4, 5 and 6)
5. How many more children in total saw the Tuesday afternoon performances than adults?

Children: $200 + 500 + 360 + 490 + 500 + 390 + 160 + 170 = 2770$

Adults: $125 + 50 + 220 + 340 + 350 + 220 + 120 + 90 = 1515$

$2770 - 1515 = 1255$